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RESEARCH ARTICLE

A Reinforcement Learning Approach to Personalized Asthma Exacerbation Prediction Using Proximal Policy Optimization

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ABSTRACT Asthma exacerbation prediction is critical for preventing severe respiratory complications and improving patient outcomes. Traditional predictive models rely on static machine learning approaches, which lack adaptability to evolving patient conditions and environmental changes. This study proposes a reinforcement learning (RL)-based model using Proximal Policy Optimization (PPO) to enhance personalized early prediction of asthma exacerbation attacks. Unlike conventional models, PPO dynamically refines its policy through iterative learning, allowing for real-time, adaptive risk assessment based on personalized patient data and environmental factors. The PPO model was trained on a comprehensive asthma dataset, integrating demographic, environmental, lifestyle, and Asthma Control Test (ACT) scores. The model achieved 96.60% accuracy, 95.79% precision, 96.65% recall, and 95.92% F1-score, outperforming baseline RL algorithms such as Deep Q-Learning (92.21% accuracy), Advantage Actor-Critic (94.34% accuracy), and Trust Region Policy Optimization (95.12% accuracy). Comparative evaluation with existing studies further demonstrated PPO's superior recall and sensitivity to exacerbation cases, making it highly suitable for personalized asthma management. RL performance metrics confirmed the model's stability and convergence, including cumulative rewards, policy, and value loss. These findings highlight PPO's potential for real-time clinical integration, enabling continuous patient monitoring and proactive intervention strategies. Future work will focus on training the model on larger, multi-source datasets to improve generalization across diverse populations. Additionally, further research will explore deployment in telemedicine, integration with wearable health technologies, and adaptability to other chronic diseases, ensuring scalability and real-world applicability in personalized healthcare systems.

INDEX TERMS Asthma exacerbation prediction, dynamic risk assessment, machine learning for asthma management, personalized health monitoring, proximal policy optimization (PPO), reinforcement learning in healthcare.

I. INTRODUCTION

Asthma is a chronic respiratory disease affecting millions worldwide, leading to recurring exacerbations that significantly impact the quality of life and, in severe cases, result in life-threatening complications [1]. Exacerbations arise from complex interactions among genetic predispositions, environmental exposures, lifestyle habits, and behavioral factors, making prediction a critical yet challenging task in clinical management [2], [3]. While artificial

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intelligence (AI) and machine learning (ML) techniques have been explored to enhance exacerbation prediction, current models often fall short in providing personalized and adaptive risk assessments due to their reliance on static clinical and environmental data sources [4], [5].

Existing asthma exacerbation prediction models primarily utilize machine learning algorithms such as Random Forest (RF), Gradient Boosting Machines (GBMs), and Support Vector Machines (SVMs). These models analyze predefined risk factors, including spirometry data [6], allergen exposure [7], wearable sensor inputs [8], and demographic information [9], to assess patient risk. While these techniques have demonstrated moderate success in population-based risk assessments, they fail to account for individual-specific triggers, such as medication adherence, comorbidities, and rapidly changing environmental conditions [10]. Furthermore, many existing models lack holistic data integration, often relying on isolated data sources rather than demographic, behavioral, and ecological factors in a comprehensive, dynamic risk assessment framework [11]. This fragmented data approach limits the ability to deliver personalized predictions, reducing the models' effectiveness in heterogeneous patient populations [12], [13].

The time-varying nature of asthma risk factors, including fluctuating air quality, weather conditions, and patient behaviors, further complicates the prediction of exacerbations [14]. Static models that do not continuously update forecasts based on evolving patient data are inherently limited in their ability to deliver real-time, personalized, clinically actionable predictions [15]. Given these challenges, there is a pressing need for adaptive models that integrate dynamic, multifactorial data sources and refine predictions in real time to improve clinical decision support and proactive patient management [12].

Reinforcement Learning (RL) presents a promising approach for addressing these challenges by enabling models to learn and optimize decision-making through interactions with their environment [16]. Unlike traditional supervised learning, RL allows continuous adaptation based on feedback-driven optimization, making it particularly well-suited for dynamic healthcare applications [17]. Among RL algorithms, Proximal Policy Optimization (PPO) has emerged as a robust and scalable solution, offering superior stability in complex, evolving datasets [18]. PPO's ability to balance exploration and exploitation enables models to refine decision-making strategies continuously, making it an ideal choice for asthma exacerbation prediction, where patient conditions evolve over time [17].

Despite its advantages, PPO has been underexplored in the context of asthma exacerbation prediction, leaving a gap in the development of adaptive, personalized predictive models capable of integrating diverse patient-specific and environmental risk factors [9], [19]. Current ML-based exacerbation prediction models fail to incorporate continuous learning mechanisms, limiting their ability to respond to real-time patient state changes [20]. This study proposes a PPO-based dynamic risk assessment model that integrates patient demographic, behavioral, and environmental exposures into an adaptive learning framework to address these limitations.

This paper presents a PPO model for predicting asthma exacerbation, with major contributions including:

1. Development of an adaptive PPO-based model that continuously learns from personalized patient data, enabling more effective prediction of asthma exacerbation.
2. Introduction of a dynamic risk assessment framework that updates predictions, incorporating evolving patient health status, demographic, behavioral, and environmental conditions.
3. A comparative analysis of the proposed PPO model against other RL techniques demonstrates its superior predictive accuracy, adaptability, and clinical applicability.

By leveraging RL's adaptive learning capability, this study aims to bridge the gap between static ML models and personalized asthma exacerbation prediction, ultimately contributing to improved patient outcomes and optimizing clinical decision support.

II. RELATED WORKS

Asthma exacerbation is a significant public health challenge, and numerous ML models have been proposed for predicting its occurrence in patients. Early research focused on statistical methods, such as logistic regression, which offered interpretable results but struggled to capture the complex non-linear relationships inherent in patient data [21]. Recent advancements in ML have led to the development of more sophisticated models, such as decision trees, SVMs, and ensemble methods like RF and GBMs, which have shown improved predictive performance [22].

A widely used method for exacerbation prediction involves ensemble learning techniques, particularly RF, which can effectively manage the high dimensionality of clinical datasets while offering robustness to overfitting [4], [23]. While Random Forest models have demonstrated strong predictive capabilities, they still exhibit limitations in capturing temporal dynamics, which are crucial in chronic conditions such as asthma [24]. Consequently, researchers have shifted toward more dynamic models, leveraging time-series and deep learning techniques such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which have proven effective in predicting exacerbations based on longitudinal patient data [25].

Despite the success of these models, a critical challenge remains, they often fail to adapt dynamically to changing patient conditions. Existing predictive models typically assume a static approach, where features are extracted at fixed time intervals without continuous updating or adaptive risk assessment [26]. This limitation calls for methods that can optimize decision-making over time, adjusting predictions as new patient data becomes available.

A. RL IN HEALTHCARE

RL has gained significant traction in healthcare for its ability to optimize sequential decision-making across various medical domains, from treatment strategies to personalized interventions. It has emerged as a promising approach, particularly for decision-making in dynamic and uncertain healthcare environments. [16]. Several studies have highlighted its application in complex medical environments, improving decision-making efficiency, resource allocation, and asthma exacerbation [27]. Zhong et al. [28] applied Deep RL to healthcare service selection within cloud computing, addressing challenges such as energy consumption, cost, and response time. Similarly, Li et al. [29] developed an RL-based algorithm utilizing electronic health records (EHRs) to optimize sequential treatment strategies for diseases like sepsis and diabetes, achieving enhanced glucose control in diabetic ketoacidosis patients. These examples underscore RL's adaptability in navigating healthcare complexities, making it a valuable tool for improving patient outcomes and operational efficiency.

In addition to optimizing treatment strategies, RL has played a key role in personalizing real-time healthcare interventions. Gönül et al. [30] introduced a dual RL model to customize mobile-based notifications that assist individuals in managing their health behaviors. Their approach, which personalizes interventions in terms of timing, frequency, and type, outperformed traditional algorithms in simulated environments, demonstrating RL's adaptability in providing personalized support. Similarly, Datta et a. [31] highlighted RL's potential in surgical decision-making, where it aids physicians in making data-driven, real-time decisions during procedures, enhancing clinical outcomes.

RL algorithms like Q-learning and Deep Q Networks (DQNs) effectively plan treatments for chronic conditions by optimizing long-term health outcomes through adaptive patient-focused policies [32]. A novel fuzzy RL model has been introduced by Su et al. [33] to personalize deep brain stimulation under communication delays, showcasing the potential of fuzzy-based RL for individualized, real-time medical interventions. Similarly, Ma et al. [34] proposed a PPO variant enhanced with random forests to solve hyperparameter optimization challenges more efficiently, highlighting the promise of hybridized RL strategies for medical and learning tasks under sparse reward settings.

In asthma management, Do et al. [35], [36] address predicting asthma attacks by incorporating risk factors such as disease severity, physiological data, behaviors, environmental triggers, and personalized scores. Do et al. [35] employs Additive Interaction Analysis with LSTM models to forecast asthma exacerbations. Do et al. [36] uses a Deep RL framework with Deep Q-learning to model future risks and enable preventative actions when risk scores exceed thresholds. Both methods utilize historical data for better asthma management. However, extensive exploration of standard RL raises practical and safety concerns, particularly in serious health conditions like asthma exacerbations, warranting reliable and safe application in healthcare.

B. PPO AND DYNAMIC RISK ASSESSMENT

PPO addresses challenges in traditional RL methods, especially in complex, real-time environments where policy stability and safety are crucial [37]. As an actor-critic method, it optimizes policies with minimal deviation from previous ones, balancing exploration and exploitation, which suits adaptive clinical decision-making with evolving patient data. Recently, PPO has been applied to healthcare, like sepsis treatment [38], showing better adaptation to patient states than traditional RL. Although not fully explored in asthma management yet, PPO's robustness and adaptability offer the potential for asthma exacerbation prediction, enabling personalized risk adjustment with real-time data.

C. LIMITATIONS IN CURRENT APPROACHES

While RL and ML models have shown promise in asthma exacerbation prediction, most rely on static feature sets and lack personalized adaptation to evolving patient data [11]. Many RL-based models are trained on retrospective datasets, limiting their applicability in real-world clinical settings where dynamic risk assessment is crucial [39]. Furthermore, the application of PPO in asthma management remains largely unexplored, leaving a gap in models that can continuously refine predictions based on patient-specific and environmental factors [9], [19]. This study addresses these limitations by proposing a PPO-based model for dynamic risk assessment, enabling personalized adaptation and improving personalized exacerbation prediction for clinical decision support.

III. METHODOLOGY

A. DATASET AND FEATURE ENGINEERING

The study uses an asthma dataset from a mobile health application called Weather Asthma (WEA) app [40], [41]. It includes 10 input variables: five weather-related and five demographic data. Weather data is real-time and location-based, while demographics are collected during registration for personalizing risk predictions. The target variable is the Asthma Control Test (ACT) score, derived from five questions scored 1 to 5, totaling 5 to 25. Lower ACT scores indicate a higher asthma risk, as per the Global Initiative for Asthma (GINA) [42]. Table 1 describes the dataset, each record has an ACT score, user demographics, and concurrent weather conditions, encompassing 2020 records with 10 inputs and one target variable. The dataset is publicly available and can be accessed from GitHub [43]. It is shared for academic and research purposes, supporting reproducibility and transparency in asthma prediction studies.

The dataset includes demographic, environmental, and clinical features. To enhance model performance, feature engineering techniques were used: Continuous variables were normalized, and Categorical variables were one-hot encoded,

TABLE 1. Asthma dataset variables.

Variable	Description
ACT_score	The total score of an ACT submitted by a user.
Temperature	Temperature value in degrees Celsius (°C).
Humidity	Humidity percentage (%) value.
Air_pressure	Air pressure value in hectopascals (hPa).
Wind_speed	Wind speed in meters per second (m/s).
UV_index	UV index range: Low to Extreme.
Location	Location of the user (city name).
Age_group	The age group of the user (18 and below, 19-30, 31-40, 41-50, Above 50).
Gender	Gender of the user (Male, Female, Other).
Outdoor_job	Frequency of outdoor job for the user (Frequently, Occasionally, Rarely).
Outdoor_activities	Likelihood of outdoor activities for the user (Likely, Neither, and Not Likely).

which assigns unique numbers (0 to n-1) to each label, maintaining consistency for repeats [44]. Temporal was structured into sequences for capturing time-dependent patterns. Six categorical variables in the asthma dataset were transformed. The dataset was divided into a training set (1414 samples, 70%) and a test set (606 samples, 30%).

B. PROBLEM FORMULATION AS A MARKOV DECISION PROCESS (MDP)

The asthma exacerbation prediction system is a dynamic framework designed to forecast exacerbations using patient data and environmental triggers. It utilizes a Markov Decision Process (MDP) to model the dynamic health and environmental conditions affecting asthma. Key MDP elements include:

i. **State S_t:** A vector at time t represents the patient's health profile and environmental data, including lung function, weather, and exposure to asthma triggers.
ii. **Action A_t:** The action space is the set of possible decisions the agent can make, such as adjusting risk predictions.
iii. **Transition Probability** $P(S_{t+1}|S_t, A_t)$**:** This indicates the likelihood of moving between states based on actions. For instance, if the agent advises against outdoor activity, this may result in better lung function next.
iv. **Reward R_t:** The reward function gives feedback to the agent, granting positive rewards for preventing exacerbations or making accurate predictions, and penalizing otherwise.

Mathematically, the goal of the MDP is to maximize the expected cumulative reward:

$$\max_{\pi} \mathrm{E}\left[\sum_{t=0}^{T} \gamma^t R_t\right] \tag{1}$$

where π is the policy, E is the expectation operator, R_t is the reward at time t, and γ is the discount factor that controls the trade-off between immediate and future rewards, and T is the time horizon.

C. PPO ARCHITECTURE

The overall system architecture shown in Figure 1 consists of an agent (PPO model) interacting with a dynamic environment representing patients' health states, environmental factors, and potential asthma triggers. The agent learns from its interaction with the environment, exploring actions and receiving rewards to optimize the prediction of asthma exacerbation events.

1) PPO AGENT MODEL

The agent predicts and acts based on the current state S_t. It learns optimal policies by exploring actions, storing memory, and receiving rewards from the environment. The PPO algorithm ensures balanced exploration and exploitation. Memory helps the agent learn action values and handle data correlations. Exploration prevents the agent from getting stuck by allowing random actions to find better policies.

2) ENVIRONMENT

The environment includes factors affecting asthma exacerbation: patient demographics and behaviors (age, gender, smoking), profile triggers (ACT score, outdoor work), and environmental conditions (weather changes like humidity, temperature, etc.). The environment gives the agent the current state, a vector of these factors, and based on the agent's action, generates a reward and updates the state for the next step.

3) STATE AND ACTION

At each time step t, the environment provides the agent with the current state S_t, representing patient demographic and environmental factors as a vector. The agent predicts if an asthma exacerbation will occur from this state.

4) REWARD FUNCTION

After each action, the agent receives a reward based on prediction accuracy: $+1$ for correct, -1 for incorrect. The goal is to maximize the expected cumulative reward $R = \sum_{t=1}^{T} \gamma^t r_t$, guided by a discount factor γ. The agent's goal is to find the optimal policy $\pi_\theta(a_t|s_t)$ that maximizes the expected cumulative reward. PPO uses a clipped surrogate objective for stable learning. The agent continuously interacts with the environment to adapt predictions to changing conditions.

D. EVALUATION METRICS

To evaluate the performance of the model, the study uses several classification metrics:

Accuracy (ACC): The percentage of correct predictions:

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{2}$$

where TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, respectively.

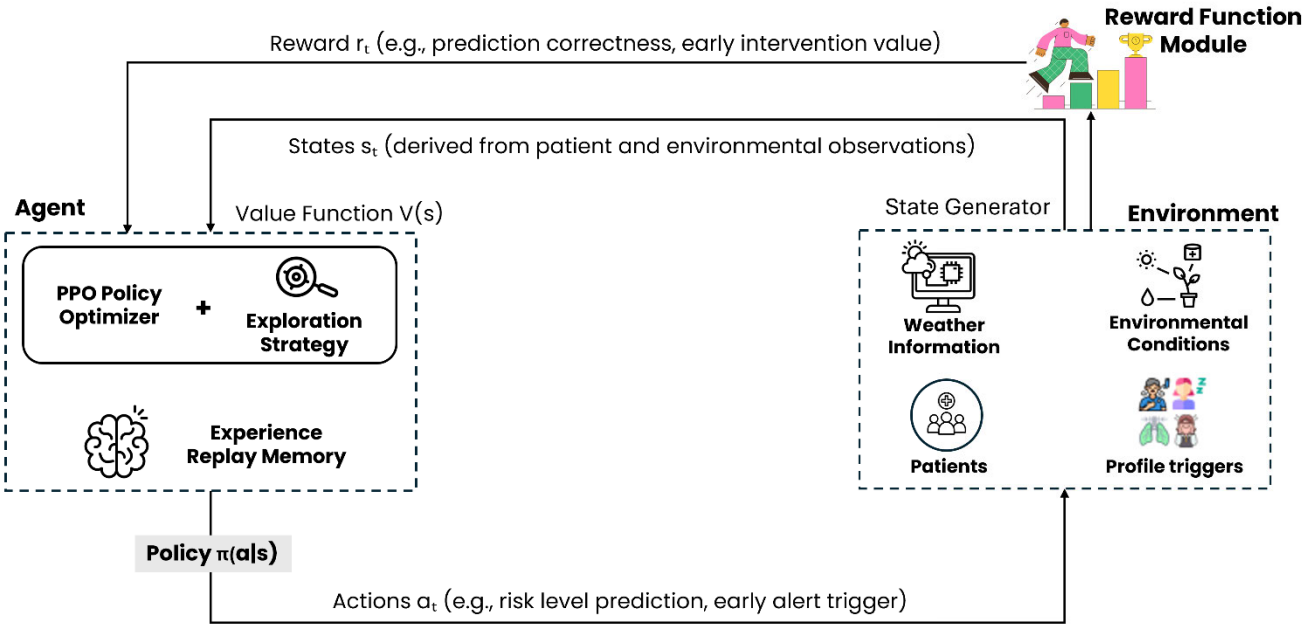


FIGURE 1. System architecture for asthma exacerbation prediction.

Precision: Measures the proportion of true positives among predicted positives:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3}$$

Recall (Sensitivity): Measures the proportion of true positives among actual positives:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4}$$

F1-score: Harmonic mean, precision and recall:

$$\text{F1} = 2.\frac{\text{Precision. Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

Receiver Operating Characteristic (ROC) is a graphical plot that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied. It plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 − Specificity).

Area Under the ROC Curve (AUC) quantifies the overall ability of the model to distinguish between classes, with a higher AUC indicating better performance. In this study, AUC represents how well the model differentiates between exacerbation and non-exacerbation cases.

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

This section outlines the computational setup and software for training and evaluating the PPO algorithm for predicting asthma exacerbations. Training occurred on a high-performance computing system, summarized in Table 2. A Tesla A100 GPU cluster on Google Cloud Platform was used for scalability. The software stack is detailed in Table 3.

TABLE 2. Hardware configuration.

Component	Specification
Processor	Intel Core i9-12900K (16 cores, 24 threads, 5.2 GHz)
GPU	NVIDIA RTX 3090 (24GB GDDR6X)
Memory (RAM)	64GB DDR5 4800MHz
Storage	2TB NVMe SSD for fast data access
Operating System	Ubuntu 22.04 LTS

B. PROCESS FOR SELECTING THE OPTIMAL PPO HYPERPARAMETERS

The selection of hyperparameters for the PPO model was carried out through a structured approach, ensuring an optimal balance between stability, computational efficiency, and predictive accuracy. The following steps outline the methods used to determine the final hyperparameter values presented in Table 4.

Step 1: Baseline Configuration and Literature Review

The initial hyperparameter values were chosen based on best practices from existing RL literature, prior studies on PPO applications in healthcare, and time-series prediction. Parameters such as learning rate, discount factor (γ), and entropy coefficient were selected based on their impact on training stability and policy convergence.

Step 2: Grid Search and Random Search for Hyperparameter Optimization

To refine the initial values, both grid search and random search methods were applied:

TABLE 3. Software stack.

Software Type	Tool/Library	Version
Programming Language	Python	3.9
Deep Learning Framework	TensorFlow	2.9.0
	PyTorch	1.13.1
RL	Stable-Baselines3 (SB3)	Latest
	OpenAI Gym	0.26.2
Data Processing	Pandas	1.5.3
	NumPy	1.24.2
Visualization	Matplotlib	3.6.3
	Seaborn	0.12.2
Optimization & Experimentation	Ray Tune	2.3.0
	Optuna	3.0.0
Virtualization	Docker	24.0.5
	Anaconda (Miniconda)	4.10.3
GPU Acceleration	CUDA	11.7
	cuDNN	8.4

a) **Grid Search**: Systematically tested different hyperparameter combinations within predefined ranges (e.g., learning rate: {0.0001, 0.0003, 0.001} and entropy coefficient: {0.005, 0.01, 0.02}).
b) **Random Search**: Randomized hyperparameter selections from a defined search space, allowing exploration beyond fixed grid values. This approach was particularly useful for batch size and clipping parameter (ε).

Step 3: Training Stability and Convergence Analysis

For each set of hyperparameters, training performance was analyzed using learning curves (reward vs. episode) to ensure:

a) **Smooth convergence**: Avoid erratic updates or premature saturation.
b) Steady improvement in reward values over multiple episodes.
c) Reduced policy variance for stable and reliable action selection.

Step 4: Sensitivity Analysis on Key Parameters

Some parameters, such as learning rate, discount factor (γ), and batch size, required fine-tuning to balance learning speed and model stability. Adjustments were made based on the following observations:

a) A learning rate of 0.0003 was chosen after testing values from 0.0001 to 0.001, as higher rates caused instability, while lower rates slowed convergence.
b) A discount factor (γ) of 0.99 was selected to maintain long-term reward optimization while avoiding short-sighted policies.
c) The batch size of 256 ensured efficient model updates without excessive memory consumption.

Step 5: Empirical Validation with Performance Metrics

Final hyperparameters were evaluated based on accuracy, precision, recall, and F1-score across validation datasets. The chosen configuration outperformed alternative setups in both generalization ability and computational efficiency.

TABLE 4. PPO hyperparameter configuration.

Hyperparameter	Value
Learning Rate	0.0003
Batch Size	256
Number of Epochs	10
Discount Factor (γ)	0.99
Clipping Parameter (ε)	0.2
Entropy Coefficient	0.01
Value Function Coefficient	0.5
Number of Steps per Update	2048

C. RESULTS

The PPO model demonstrated 96.60% accuracy in predicting asthma exacerbations using the test dataset. It balanced false positives and negatives effectively, with a precision of 95.79% and a recall of 96.65%. The F1-score was 95.92%, indicating overall reliability, though there is room to enhance sensitivity for rare cases and precision for borderline situations. The PPO agent's learning and policy optimization were evaluated using key RL metrics: reward, value loss, and policy loss across episodes. Table 5 shows the top 10 episodes by reward.

TABLE 5. Top 10 episodes from PPO training.

Rank	Episode	Reward	Value Loss	Policy Loss
1	162	602	0.3061	0.0688
2	119	590	0.2785	0.0261
3	171	586	0.2983	0.0337
4	118	586	0.2576	0.0374
5	180	580	0.1983	0.1031
6	154	582	0.3398	0.0152
7	112	578	0.2706	0.0060
8	163	576	0.3150	0.0126
9	114	576	0.2291	0.0444
10	94	574	0.2128	0.0137

The reward values reflect total rewards per episode, with higher values signifying better policy learning. The top episode (Episode 162) earned the highest reward of 602, showcasing effective long-term outcome optimization. The value loss measures estimation errors, with lower values indicating better predictions. Top-ranked episodes had lower value loss (0.1983 to 0.3398), indicating stable learning. The policy loss measures update magnitude, with lower values implying stability. Episode 112 had the lowest policy loss (0.0060), while Episode 180 had the highest (0.1031), indicating larger adjustments.

Figure 2 shows the PPO model's learning progression. Initially, the agent's negative rewards reflect a suboptimal policy and limited environmental understanding. Rapid cumulative reward increase occurs in the first 25 episodes, indicating quick learning and decision-making improvement. Between episodes 25 and 100, the learning curve grows steadily with slight fluctuations, as the agent refines its policy and adapts to the environment. After episode 100, the reward stabilizes around 600, signifying model convergence to an optimal policy. Further training yields marginal improvements, with slight variations suggesting ongoing but minor policy refinements.

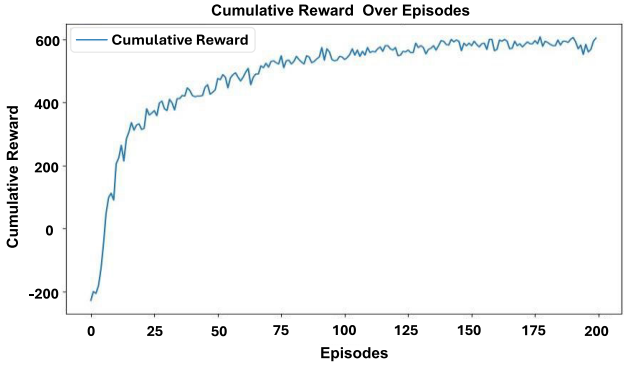


FIGURE 2. Cumulative reward progression over training episodes.

Figure 2 shows that the PPO-based model effectively learns an optimal policy for predicting asthma exacerbations. The stable cumulative reward indicates robust and efficient RL, which ensures accurate predictions and effective decision-making.

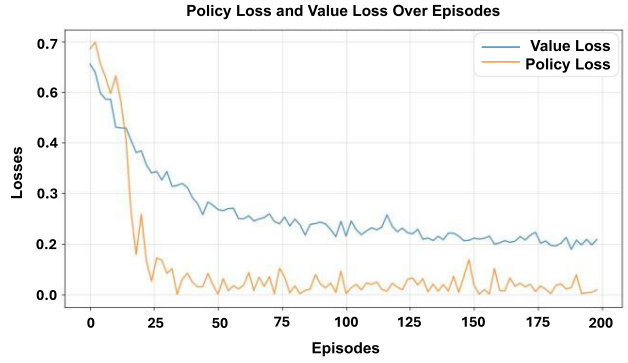


FIGURE 3. Policy loss and value loss during PPO training.

The trends of policy loss and value loss over training episodes, Figure 3, reveal the model's learning and convergence. Initially, both losses are high, indicating major policy updates and value instability. A sharp decline within the first 25 episodes signals rapid policy improvement. After episode 25, policy loss nears zero, showing stabilization and optimal policy learning by the PPO agent. Value loss also drops, though it stays slightly higher, as the value function continues refining predictions. Beyond episode 100, both losses stabilize with minor fluctuations, confirming convergence and stable performance. Occasional value loss spikes imply small value function adjustments, expected during long-term reward refinements. The trends confirm effective training, minimizing policy and value loss, achieving a stable RL model for asthma exacerbation prediction, and demonstrating the model's generalization and stability.

The trends observed in Figures 2 and 3 are further substantiated by the top-performing episodes summarized in Table 5. When comparing these episodes, it becomes evident that the PPO model consistently maintains high cumulative rewards while balancing stability and efficiency. For instance, Episodes 119, 118, and 171 show similar reward levels to Episode 162 but with varied policy and value loss dynamics, indicating that PPO effectively adapts its policy updates across different training conditions. Notably, Episode 180, despite having the highest policy loss among the top 10, still achieved a high cumulative reward, highlighting PPO's ability to make impactful adjustments without compromising overall performance. The consistency in value loss across episodes (ranging from 0.1983 to 0.3398) reflects steady value estimation, while low policy losses across most episodes suggest that the model quickly converged to an optimal policy. This episode-level comparison reinforces the broader training trends and confirms that PPO offers a robust, stable, and high-performing RL strategy for personalized asthma exacerbation prediction.

The ROC curve in Figure 4 evaluates the PPO model's ability for asthma exacerbation prediction, showing the balance between true positive and false positive rates. AUC of 0.96 reflects strong discrimination. A near-steep rise confirms high true positives with low false positives. This high AUC supports metrics like accuracy (96.60%), precision (95.79%), recall (96.65%), and F1-score (95.92%), validating the effectiveness of RL-based optimization for reliable asthma exacerbation risk prediction.

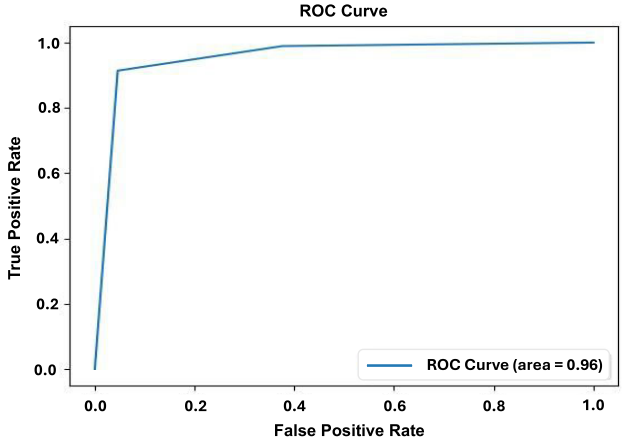


FIGURE 4. ROC curve depicting model performance for asthma prediction.

D. BASELINE MODELS

The PPO model shows strong predictive performance compared to other RL algorithms, as shown in Table 6 and Figure 5. It achieved 96.60% accuracy, surpassing DQL (92.21%), A2C (94.34%), and TRPO (95.12%), but slightly lower than SAC (97.43%). This suggests PPO effectively learns optimal policies for predicting asthma exacerbations while generalizing well. The PPO, DQL, A2C, TRPO, and SAC models were trained and evaluated on the Weather Asthma App (WEA) dataset using a uniform preprocessing pipeline and a 70-15-15 split for training, validation, and testing. Hyperparameters were optimized through standard configurations and grid search.

TABLE 6. Comparison of RL algorithms based on evaluation metrics.

Algorithm	Accuracy	Precision	Recall	F1-Score
Proposed PPO	**96.60**	**95.79**	**96.65**	**95.92**
DQL	92.21	90.91	91.87	91.38
A2C	94.34	93.67	94.01	93.82
TRPO	95.12	94.04	95.21	94.61
SAC	97.43	96.21	96.09	96.65

PPO has a precision of 95.79%, higher than DQL and A2C, but slightly lower than SAC's 96.21%. Its recall of 96.65% is the highest among all models, indicating effective identification of actual exacerbation cases and reduction of false negatives, which is crucial in healthcare. PPO's F1-score of 95.92% shows a balance of precision and recall, making it a strong choice for asthma exacerbation prediction. Although SAC slightly surpasses PPO in accuracy, precision and F1-score, PPO's excellent recall highlights its advantage in sensitivity to exacerbation cases. Overall, PPO offers a strong balance of performance in predicting asthma exacerbations. Additionally, PPO's training efficiency and stable convergence behavior further support its suitability for real-time clinical deployment, where both accuracy and reliability are essential.

The proposed PPO model performance was qualitatively compared based on the results reported in previous studies on asthma exacerbation prediction, as summarized in Table 7. As these studies may have used different datasets, feature sets, and experimental configurations, the comparison is intended to offer contextual insight rather than serve as a strict performance benchmark.

Table 7 shows that the PPO model is competitive, with 96.60% accuracy, comparable to [46] (96.52%) and [51] (98.57%). It surpasses models like [45] (90.60%), [47] (94.37%), [48] (87.10%), and [50] (78.43%) in predicting asthma exacerbations. PPO's precision is 95.79%, slightly less than [46] (96.52%) and [51] (97.87%), but better than [47] (94.88%) and [50] (77.94%). Its recall is 96.65%, closely matching [46] (96.52%) and [51] (97.91%), and exceeding [45] (70.2%) and [50] (79.41%). The F1-score

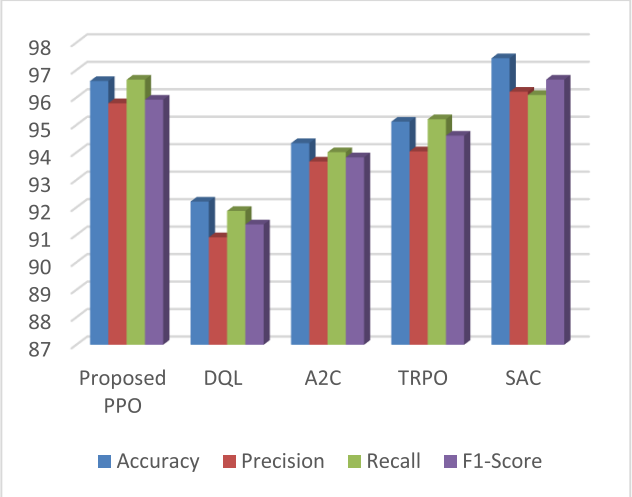


FIGURE 5. Comparison of RL algorithms based on evaluation metrics.

TABLE 7. Potential applications of PPO-based framework vs. existing models.

Ref.	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
[45]	90.60	90.91	70.2	-
[46]	96.52	96.52	96.52	96.52
[47]	94.37	94.88	95.14	94.96
[48]	87.1	97.9	82.5	-
[49]	73.8	68.7	72.7	68.9
[2]	-	93	90	-
[50]	78.43	77.94	79.41	71.05
[51]	98.57	97.87	97.91	98.93
PPO	**96.60**	**95.79**	**96.65**	**95.92**

of PPO (95.92%) indicates a strong balance of precision and recall, making it a reliable predictor. Although [51] performed slightly higher across all metrics, and PPO is more effective than many other studies in minimizing false positives and negatives.

V. DISCUSSION

The Proposed PPO model shows significant improvement in predicting asthma exacerbations, offering better accuracy, sensitivity, and stable convergence than traditional methods. It achieves a predictive accuracy of 96.60%, outperforming models like DQL, A2C, and TRPO. Its ability to optimize decisions results in an F1-score of 95.92%, balancing precision and recall. The high recall of 96.65% surpasses other models, crucial for reducing false negatives in clinical settings and preventing severe health risks.

The cumulative reward trend in Figure 2 demonstrates PPO's efficiency, with rapid learning and convergence after 100 episodes. Unlike traditional methods prone to instabil-

ity or overfitting, PPO's clipped surrogate objective enables smoother policy updates and stable learning. Trends in policy and value loss shown in Figure 3 further confirm this convergence, reflecting minimized losses and robust policy optimization.

While some external studies, such as [51] achieved slightly higher with accuracy of 98.57%, precision of 97.87%, recall of 97.91%, and an F1-score of 98.93% using a supervised hybrid model that combines Disperse Foraging Salp Swarm Algorithm (DFSSA) for feature optimization and Kernel Extreme Learning Machine (KELM) for classification, PPO's recall and balanced F1-score values establish it as a strong and dependable model for clinical decision-making. Its superior performance over DQL, A2C, and TRPO confirms its advantage in RL settings for medical prediction tasks. PPO stands out for its accuracy, recall, and learning stability, making it a strong candidate for real-time asthma exacerbation prediction across diverse patient populations.

A. IMPLICATIONS FOR HEALTHCARE

Integrating adaptive RL models like PPO in healthcare, particularly for predicting asthma exacerbations, enhances personalized patient management. Static models can't adjust to individual differences, but PPO's dynamic learning adapts to evolving patient data, enabling early intervention and tailored treatments by considering environmental and physiological factors. PPO-trained models identify high-risk patients early, cutting hospital visits and costs while improving outcomes. With a high recall of 96.65%, PPO ensures critical exacerbations are detected, supporting clinical decisions with real-time alerts for preventive actions. PPO holds potential in remote patient monitoring and wearable tech, aligning with modern healthcare's shift to digital solutions for continuous monitoring of chronic diseases. It processes real-time data for personalized care, enhancing patient autonomy and offering providers actionable insights. Unlike traditional models requiring manual tuning, PPO offers scalable continuous learning and policy adjustment, improving predictive accuracy and facilitating proactive interventions in asthma management. By handling real-time data and tailoring treatments, PPO reduces exacerbation risks, highlighting its transformative role in clinical decision support.

B. LIMITATIONS

The PPO model performs well in predicting asthma exacerbations, but it has limitations in real-world clinical settings concerning data availability, scalability, computational demands, and interpretability. Its effectiveness relies on large, high-quality datasets that depict patient variability. Access to comprehensive asthma-related records is challenging due to privacy issues and incomplete data, and existing datasets may have a class imbalance, impacting model generalization despite strategies like Synthetic Minority Over-sampling Technique (SMOTE). PPO's computational expense also poses a challenge, requiring significant GPU resources for training. While High-Performance Computing (HPC) clusters aid during training, deploying the model in resource-limited settings like rural clinics or mobile health apps is impractical without model compression. Additionally, deploying in real-time clinical workflows risks latency due to the model's need for continuous interaction with an environment.

Model interpretability is challenging as deep RL models like PPO operate as black boxes, complicating understanding for healthcare professionals. Explainable AI (XAI) methods like SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) offer some interpretability, but more transparency is needed for clinical use, causing hesitation in adopting AI in healthcare. Real-world adaptability is another issue; PPO was trained on historical data, yet actual conditions vary with new health factors and environmental changes. Continual learning and retraining are needed to maintain accuracy, or else prediction effectiveness declines. Regulatory and ethical hurdles also challenge AI in clinical settings as models must meet standards such as HIPAA, GDPR, and FDA, requiring validation and testing that are time-consuming and expensive.

C. FUTURE WORK

To enhance the PPO model for predicting asthma exacerbation, future improvements focus on model refinement, clinical integration, adaptation to chronic diseases, and real-time deployment. Integrating PPO with clinical decision support systems (CDSS) and embedding it into electronic health records (EHR) would allow continuous monitoring of patient data for personalized treatment, risk alerts, and early intervention. Using wearable health devices could improve accuracy with real-time data. Expanding PPO to chronic diseases like Chronic Obstructive Pulmonary Disease (COPD), cardiovascular, and diabetes would prevent severe complications and create a scalable AI-driven health system. Enhancing interpretability through XAI techniques such as attention mechanisms and feature attribution (e.g., SHAP, LIME) will make decisions more transparent, boost physician trust, and aid clinical adoption.

Improving the computational efficiency and real-time inference speed of PPO models is essential, especially for deployment in low-resource or mobile health settings. Future work should focus on model compression techniques like quantization, pruning, and knowledge distillation to create lightweight versions, maintaining high performance with reduced processing power. Continual learning and adaptability are crucial to keep models effective as patient conditions and healthcare protocols change. Online learning can help PPO update with new patient data, keeping predictions relevant. Clinical validation and regulatory compliance, including HIPAA, GDPR, and FDA standards, are critical for real-world implementation. Large-scale trials and validations are necessary to ensure generalizability and robustness across diverse populations, enabling the transition from research to practical healthcare applications.

VI. CONCLUSION

This study introduced a PPO-based model for early prediction of asthma exacerbation, showing high accuracy and stability. It surpassed baseline RL approaches with 96.60% accuracy, 95.79% precision, 96.65% recall, and 95.92% F1-score, balancing sensitivity and specificity. PPO refines decision-making through RL, effectively identifying risks while maintaining stable training. The research introduced a dynamic risk assessment using real-time data, allowing personalized management by adapting to individual conditions and historical patterns. This is crucial for timely intervention in clinical settings. Integrating PPO with patient monitoring and EHRs enhances early warnings, helping physicians adjust treatments proactively. The study emphasizes RL's potential in predictive healthcare, reducing hospitalizations, improving quality of life, and assisting healthcare providers. Further research should focus on real-world applications, compliance, and wearable tech integration to improve PPO models. Advancements in RL, like PPO, can revolutionize predictive healthcare, enabling early detection and personalized interventions for better asthma patient outcomes.

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